HUJI · 67594 · INTRODUCTION TO COMMUNICATION NETWORKS

Week 02

Layering; Multiple Access on a Shared Link

modularity · protocols · the five layers · encapsulation · TDMA/FDMA · ALOHA · CSMA/CD · Ethernet · CSMA/CA · RTS/CTS

What this week contains. This week's only file is the lecture ("2 layers+broadcast", 103 slides); there was no separate tutorial file. So Part A is the lecture, written as a textbook chapter, followed by practice problems and a cheat sheet. The lecture's "Probability 101" block (slides 49–67) repeats Tutorial 1, which Week 01 Part B solves in full. Here it gets a short recap, plus the one new mathematical tool that the MAC analysis needs, the limit $(1 - 1/N)^N \to 1/e$.

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PART A · LECTURE 2

Layering and the Broadcast Link

SECTION 01

Modularity: why networks are built in pieces

A network is one of the largest systems ever built. Nobody could design it as a single program. Like every large computer system, it is split into *modules*.

DEF · MODULE, INTERFACE, ABSTRACTION

- A module is a part of a system that does one well-defined job.
- Its interface is the list of things other modules may ask it to do, and what they get back.
- An abstraction is what the interface lets you *forget*: you use the module without knowing how it works inside.

EXAMPLE · MODULARITY YOU ALREADY USE

- A **library**: you call `sort(list)` without knowing whether it runs quicksort or mergesort.
- A **programming language and its compiler**: you write `x = y + z` without knowing the CPU's registers and instruction set. The same program compiles for an Intel laptop and an ARM phone.

WHAT WELL-DEFINED INTERFACES BUY YOU

- **Hidden implementation**: a module's inside can be changed freely, and the change stays local.
- **Extensibility**: new functionality is added by adding new modules.
- **Evolution**: the system can keep improving piece by piece, for decades.
- **Isolated assumptions**: each module's assumptions stay inside it, and only high-level abstractions are shown outside.

PITFALL · MODULARITY IS NOT FREE

Hiding information can hurt performance. A module that cannot see inside its neighbours cannot use what they know. For example, a module that does not know the link is wireless may treat a loss as congestion. The trade-off is **ease of implementation vs. performance**, and networking chooses ease almost every time.

> 1.1 What is special about *network* modularity

In a single program all modules run on one machine. In a network the implementation is **distributed across many machines**, hosts and routers. So the designers must decide three things:

1. **How to break the system into modules**: the answer is layering (this lecture).
2. **Where each module is implemented**: at the hosts, or inside the network?
3. **Where state is stored**: the principle of fate-sharing (later in the course).

The lecture addresses each choice when it becomes relevant; this week we meet the first two.

SECTION 02

Layering

>2.1 The tasks, bottom up

To move data from one application to another, the network has to solve these tasks, from the physical bottom to the top:

1. electrons (or light, or radio waves) on the wire;
2. bits on the wire;
3. packets on the wire;
4. deliver packets across a **local** network (a LAN), using **local addresses**;
5. deliver packets across **many** LANs, using **global addresses**;
6. make sure the packets get there (for example, handle loss);
7. do something useful with the data.

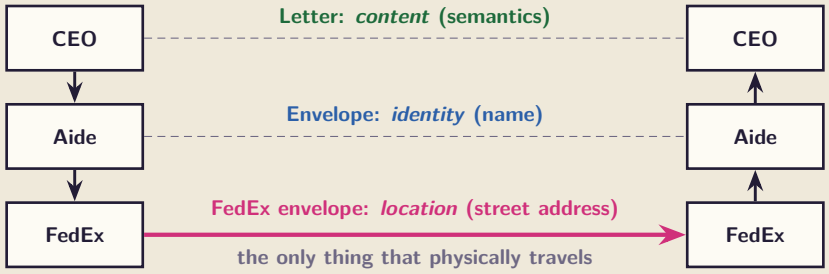
DEF · LAYERING

Layering organizes the tasks into a stack of layers. It is a particularly **strict form of modularity**: each layer interacts only with the layer directly above it and the layer directly below it. A layer uses the service of the layer below and offers a service to the layer above.

>2.2 Layering in the real world: the CEO's letter

EXAMPLE · A LETTER BETWEEN TWO COMPANIES

1. CEO A writes a letter to CEO B, folds it and hands it to an administrative aide.
2. The aide puts the letter in an envelope with CEO B's name on it and takes it to FedEx.
3. The FedEx office puts the envelope in a larger FedEx envelope, writes the company's name and street address on it, and puts it on a truck.
4. FedEx delivers it to company B. There, the FedEx envelope is opened, the aide reads the name and passes the inner envelope to CEO B, and CEO B reads the letter.



The path of the letter (slide 10). Each dashed line joins two *peers* that understand the same thing. The real movement goes down on the left, across at the bottom, and up on the right.

THREE LESSONS OF THE LETTER

- **Peers understand the same things, and no one else does.** The FedEx driver never reads the letter; the CEO never reads the street address.
- **Each layer adds its own wrapping**, so the lowest layer carries the most packaging.
- **Each layer only talks to its neighbours**: the CEO deals with the aide, never with the truck.

> 2.3 The resulting five layers

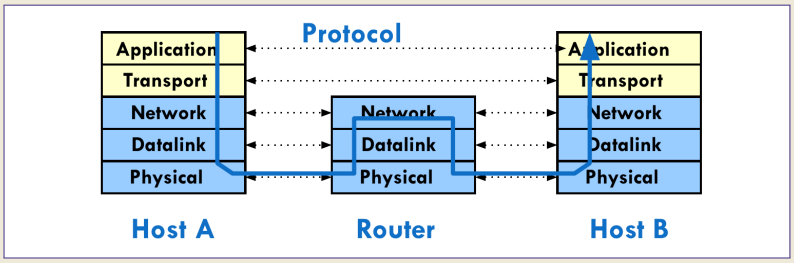
Each task becomes a layer. Two tasks are folded into the layer above them.

LAYER	TASK	WHAT IT IS ABOUT
5 Application	do something with the data	web, e-mail, video, games
4 Transport	ensure the packets get there	between *processes*; reliability
3 Network	deliver across many LANs	**global** addresses; routing
2 (Data) Link	deliver across one LAN (incl. packets on wire)	**local** addresses; sharing the medium
1 Physical	bits on the wire (incl. electrons)	voltages, light, radio

A MNEMONIC FOR THE FIVE

Read from the bottom: **P**hysical, **L**ink, **N**etwork, **T**ransport, **A**pplication. Each layer answers one question: *how is a bit sent? to which neighbour? to which machine in the world? to which program on it? and what does the program do with it?*

> 2.4 Who implements which layers?



Slides 13–15. Host A and Host B implement all five layers; the router in the middle implements only the bottom three. The blue line is the *physical* path of the data: down Host A's stack, across the wire, up to the router's network layer, down again, and up Host B's stack.

THE TOP TWO LAYERS LIVE ONLY AT THE HOSTS

Routers implement **physical, link and network**. **Transport and application are implemented only at the end hosts**. Inside the network, nobody looks at the transport or application data. (This is the answer to the second design question of §1.1: *where* each module is implemented.)

DEF · LOGICAL VS. PHYSICAL COMMUNICATION

- Logical communication: each layer behaves as if it talks directly to its peer, the same layer on the other machine. Host A's transport layer "talks to" Host B's transport layer. The router's network layer "talks to" both hosts' network layers.
- Physical communication: the bits really go *down* to the physical layer, travel peer to peer across each link, and go back *up* to the relevant layer.

SECTION 03

Protocols

DEF · PROTOCOL

A protocol is an **agreement on how to communicate**: to exchange data, and to coordinate the sharing of resources. A protocol specifies:

- its syntax: how the protocol is structured, meaning the format of the messages and the order in which they are sent and received;
- its semantics: what the bits mean, meaning how to respond to each message or event.

EXAMPLE · PROTOCOLS IN LIFE

- **Asking a question in class**: raise a hand, wait to be called on, then speak.
- **Turn-taking in conversation**: a pause is the signal that the next person may respond.
- **Boarding and leaving an airplane**: by rows or zones. Not all countries use the same protocol!
- **Answering the phone**: you say "hello" as the signal for the caller to talk; the caller identifies themselves, and then the conversation starts.

Each one has syntax (what is said, in what order) and semantics (what each signal means and what to do next).

EXAMPLE · HTTP, AN APPLICATION-LAYER PROTOCOL

A browser asks a web server for a page. The request (syntax: a method, a path, a version, then header lines, then an empty line):

```
GET /courses/archive/spring06/cos461/ HTTP/1.1
Host:  www.cs.princeton.edu
User-Agent:  Mozilla/4.03
(empty line)
```

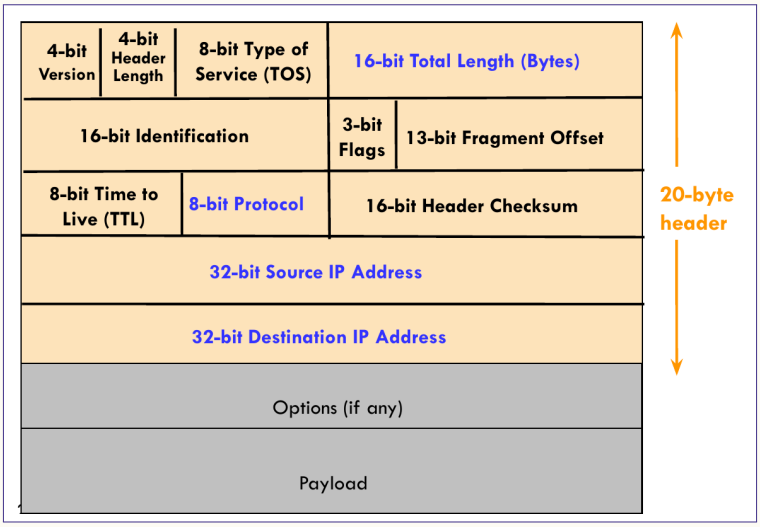
The response (semantics: `200 OK` means "here it is"):

```
HTTP/1.1 200 OK
Date:  Mon, 6 Feb 2006 13:09:03 GMT
Server:  Netscape-Enterprise/3.5.1
Last-Modified:  Mon, 6 Feb 2006 11:12:23 GMT
Content-Length:  21
(empty line)
Site under construction
```

`Content-Length:  21` tells the browser exactly how many bytes of body to read: "Site under construction" is 23 characters, so the slide's example is slightly off, but the idea is that the header announces the length.

EXAMPLE · IP, A NETWORK-LAYER PROTOCOL

Lower layers use binary formats with fixed-size fields. The IP packet header:



Slide 20. Each row is 32 bits. Without options the header is 5 rows = 20 bytes. The fields you need now: *Version* (4 for IPv4), *Total Length*, *Time to Live* (decremented by each router; the packet is dropped at 0), *Protocol* (which transport protocol is inside, e.g. TCP), and the 32-bit *source* and *destination* addresses.

> 3.1 Standardization

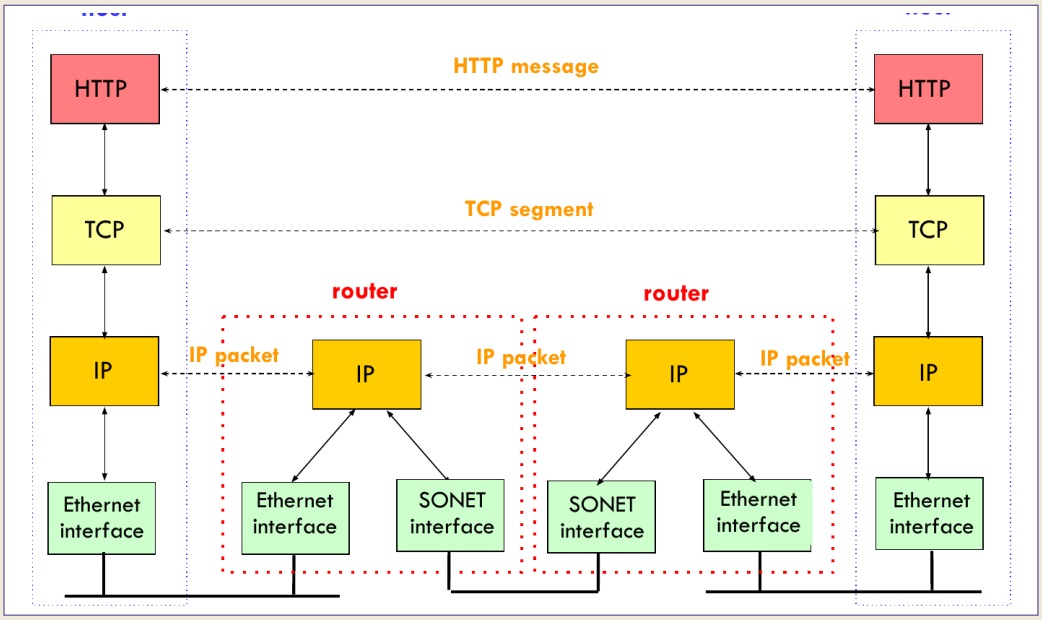
WHY STANDARDS

Everyone must follow the same protocol. Very small changes can make a big difference, or stop communication altogether: if one side thinks the length field is 16 bits and the other 32, nothing works. Hence **standards**. A standard fixes the protocol, *not* the code, so there can be many independent implementations (Windows, Linux and macOS each have their own TCP).

DEF · IETF AND RFCS

The Internet Engineering Task Force (IETF, `ietf.org`) writes the Internet's standards. It works in *working groups* focused on specific issues, and publishes its documents as numbered Requests For Comments (RFCs), archived at `rfc-editor.org`. (Link-layer standards such as Ethernet and WiFi come from the IEEE 802 committees.)

> 3.2 The Internet protocol suite, end to end

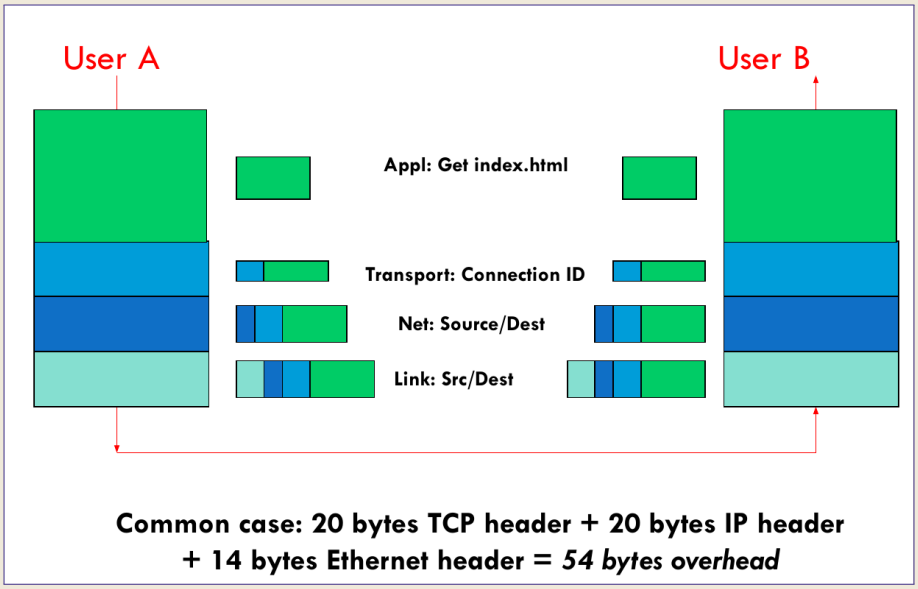


Slide 22. Each layer's unit of data has its own name: the application sends an *HTTP message*, the transport layer a *TCP segment*, the network layer an *IP packet* (datagram), and the link layer a *frame*. The two routers process only IP and below. Notice that the link changes along the way: Ethernet, then SONET, then Ethernet again. IP runs over all of them.

> 3.3 Encapsulation

DEF · ENCAPSULATION

Going down the stack, each layer takes what it received from above as its *payload* and adds its own *header* in front. This is encapsulation. Going up the stack at the receiver, each layer removes its own header and passes the payload up (decapsulation).



Slide 23. User A's request "Get index.html" (green) gets a transport header (connection ID), then a network header (source and destination addresses), then a link header (local source and destination). At User B the headers are removed in the opposite order.

Step by step for one web request [KR 1.5.2]:

1. The application builds the HTTP message M.
2. Transport adds a header H_t (which process on which host): segment $= H_t\, M$.
3. Network adds H_n (global source and destination addresses): datagram $= H_n\, H_t\, M$.
4. Link adds H_l (local source and destination on this LAN): frame $= H_l\, H_n\, H_t\, M$.
5. The physical layer sends the bits. Each router strips H_l, reads H_n, and wraps the datagram in a *new* link header for the next link.

EXAMPLE · HOW MUCH IS OVERHEAD?

Common case: 20 bytes TCP + 20 bytes IP + 14 bytes Ethernet = 54 bytes of headers.

- A full frame with 1460 bytes of data: $1460/(1460+54) = 96.4\%$ of the bytes are useful.
- A single typed keystroke (1 byte of data, e.g. in a remote shell): $1/55 = 1.8\%$ useful. Mice pay heavily for headers.

PITFALL · 54 BYTES COUNTS HEADERS ONLY

The 54 bytes are the three *headers*. On the wire, Ethernet also adds a 4-byte CRC *trailer* after the data (and an 8-byte preamble before the frame) [KR 6.4.2]. So the real per-frame overhead is a bit larger.

SECTION 04

Characterizing the layers

DEF · SERVICE, INTERFACE, PROTOCOL (OSI TERMS)

Every layer is described by three things:

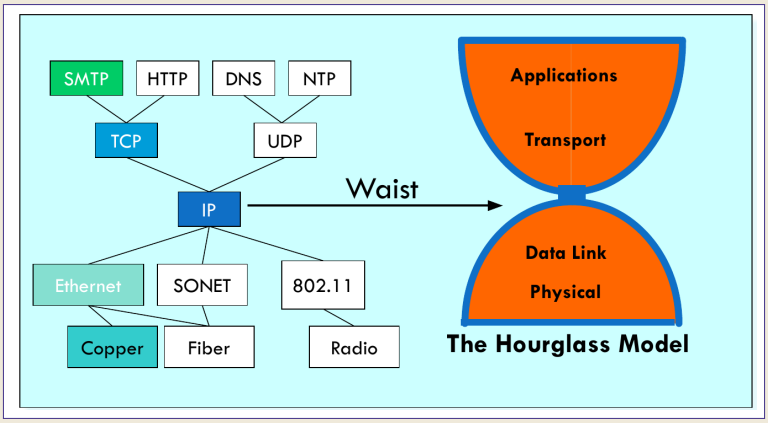
- its service: *what* the layer does for the layer above;
- its service interface: *how* the layer above accesses the service;
- its protocol: how the *peers* on different machines communicate to provide the service. The protocol governs what happens *between* machines, not how the layer is coded on one machine.

A layer can have **many implementations**, and this is what allows innovation.

LAYER	SERVICE	PROTOCOLS DO	EXAMPLES
1 Physical	move bits between two systems on *one* physical link	coding of bits: voltage levels, bit duration	optical, wireless (not our focus)
2 Link	exchange frames with hosts on the same LAN, using local addresses	Media Access Control (MAC)	Ethernet, 802.11, Frame Relay, ATM
3 Network	deliver packets to any destination, across many layer-2 networks, using global addresses	routing	IP (only!)
4 Transport	communication between *processes*; perhaps reliability, rate adaptation	reliability, flow control, splitting large messages	TCP, UDP
5 Application	any service to the end user	depends on the application	HTTP, SMTP, Skype, BitTorrent

- **Link interface**: "send this frame to that local host"; "deliver to me the frames addressed to me". The link layer is not limited to a single wire: a LAN may have many hosts.
- **Network interface**: "send this packet to that global destination"; "deliver to me the packets for this host". "Inter-network" means across *different* layer-2 networks, e.g. from Ethernet to 802.11.
- **Transport interface**: "send this message to that process on that host"; a local process receives the messages sent to it. Multiplexing here means that many processes on one host share the host's network connection.

> 4.1 The narrow waist



The Internet hourglass (slide 30). Many applications at the top, many link and physical technologies at the bottom, and exactly one protocol in the middle: IP.

WHY ONLY ONE NETWORK-LAYER PROTOCOL

The slide asks: "IP: why no other examples?" Because **IP is the narrow waist**. Every application runs over IP, and IP runs over every link technology.

- A new application (top) only has to work with IP, and it runs on every network.
- A new link technology (bottom) only has to carry IP, and every application works over it.

With A applications and L link technologies, a single waist needs $A + L$ adaptations instead of $A \times L$. With 100 applications and 20 links: 120 instead of 2000. The narrow waist is what makes interoperability possible. The price: changing IP itself is extremely hard (IPv6 has been "coming" for decades).

PITFALL · "WASTE" VS. "WAIST"

Slide 30's title says "The Narrow *Waste* of the Internet". The word is **waist**, as in the narrow middle of an hourglass, which is what the figure shows.

SECTION 05

Building the Internet bottom up: the link layer

The course now builds the Internet from the bottom. **Challenge I: communicating within a LAN.** For now we look only at the physical and link layers (the physical layer itself is not the focus of the course). Transport and network come later; the link layer is needed to interconnect LANs.

> 5.1 Terminology

DEF · NODE, LINK, FRAME

- Nodes: end hosts *and* relays (routers, switches): anything that runs a link layer.
- Links: the communication channels that connect *adjacent* nodes along the path. They can be wired, wireless, or logical (a broadcast domain).
- Frame: the layer-2 packet. It *encapsulates* a network-layer datagram.

The **data-link layer** is responsible for transferring a datagram from one node to an *adjacent* node over one link.

EXAMPLE · THE TRIP FROM HONOLULU TO JERUSALEM [KR 6.1]

A taxi from Honolulu to HNL airport, a plane to Newark (EWR), a plane to Tel Aviv (TLV), a train to Jerusalem.

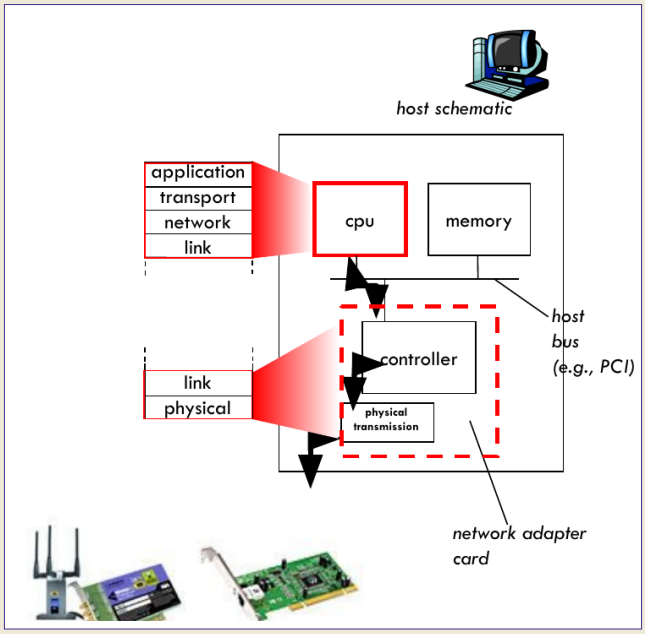
TRAVEL	NETWORK
passenger	datagram
each segment of the trip	one link
transportation mode (taxi, plane, train)	link-layer protocol
travel agent who plans the route	routing algorithm

The datagram crosses different links with *different* link protocols (Ethernet on the first link, 802.11 on the next), and each one provides different services. A taxi does not promise what a plane does, and some links are reliable while others are not.

> 5.2 Services of a single link

- **Link access and addressing.** When the channel is a shared medium, someone must decide who sends. MAC addresses in the frame header identify the source and the destination. They are different from IP addresses (§16).
- **Reliable delivery between adjacent nodes** (error detection and correction, retransmissions). It is seldom used on low-bit-error links (fibre, some twisted pair), but needed on wireless links, which have high error rates. This is revisited later in the course.
- "...?": the slide leaves the list open. [KR 6.1.1] also lists *framing* (wrapping the datagram) and *error detection* as separate services.

>5.3 Where is the link layer implemented?



Slide 39. Inside a host, application, transport and network run in software on the CPU. Link and physical are in the *network adapter*.

DEF · NETWORK INTERFACE CARD (NIC)

The link layer runs **in every host**, in an adapter, also called a network interface card (NIC): e.g. an Ethernet card or an 802.11 card. The NIC implements the link and physical layers. It attaches to the host's system bus and is a combination of hardware, software and firmware.

ADAPTERS COMMUNICATING

- **Sending side**: encapsulates the datagram in a frame and adds error-checking bits (and, if used, reliable-delivery machinery).
- **Receiving side**: checks for errors, extracts the datagram and passes it up to the network layer.

SECTION 06

The multiple access problem

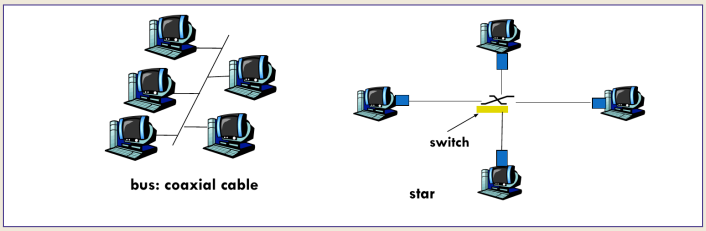
DEF · BROADCAST CHANNEL AND COLLISION

On a single shared broadcast channel, every frame sent by one node is received by all the others. If a node receives two or more signals at the same time, there is a collision: the signals are tangled together and all the colliding frames are lost, and the channel was wasted during the collision [KR 6.3].

DEF · MULTIPLE ACCESS PROTOCOL

A multiple access protocol (MAC protocol) is a **distributed algorithm** that determines how the nodes share the channel, i.e. *when* each node may transmit. The catch: the communication about how to share the channel must use the channel itself. There is no separate "out-of-band" channel for coordination.

> 6.1 Broadcast domains: wired



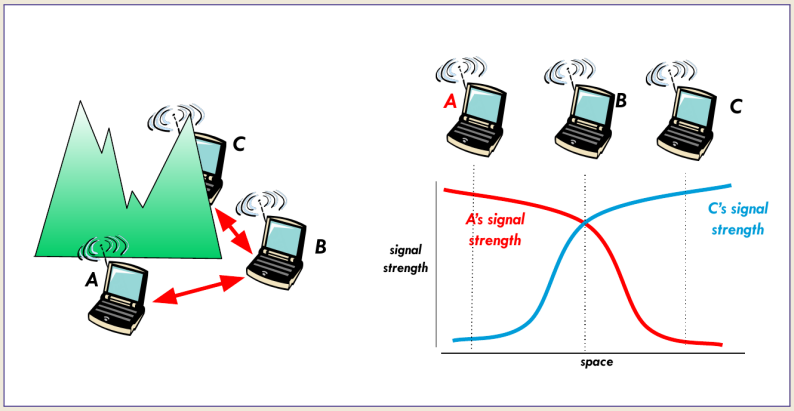
Slides 42, 96. Left: a *bus*, one coaxial cable shared by every node. Right: a *star*, every node on its own cable to a switch.

- **Bus** (popular until the mid-1990s): all nodes are in the same collision domain: any two of them can collide.
- **Star** (today): an *active switch* sits in the centre. Each node runs a separate Ethernet with the switch, so nodes do not collide with each other. Switches come later in the course.

PITFALL · COLLISION DOMAIN VS. BROADCAST DOMAIN

Slide 42 says the bus nodes are in the same "collision domain"; slide 96, the same slide repeated, says "broadcast domain". On a bus the two coincide, so both are true. In general they are different things: a *collision domain* is the set of nodes whose transmissions can collide; a *broadcast domain* is the set of nodes that receive a broadcast frame. The switched star still has one broadcast domain, but each cable is its own collision domain.

> 6.2 Broadcast domains: wireless



Slide 43. Left: A and C are separated by a mountain; both reach B, but they cannot hear each other. Right: A's signal (red) fades with distance; so does C's (blue). At B both are strong enough; at A, C's signal is too weak to sense, and vice versa.

DEF · HIDDEN TERMINAL AND FADING [KR 7.2]

In wireless, "everyone hears everyone" is not true.

- Hidden terminal problem: A and C both reach B, but an obstacle hides them from each other. Neither can tell that the other is transmitting, so they collide at B.
- Fading: signal strength decreases with distance. Even without an obstacle, A and C may be too far apart to hear each other, although B, in the middle, hears both.

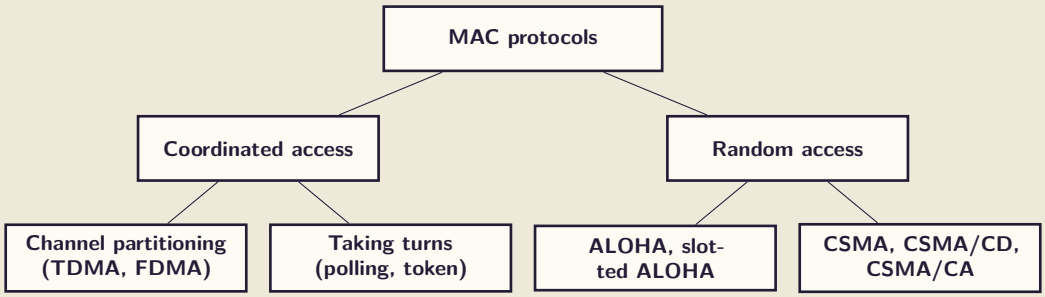
> 6.3 The ideal multiple access protocol

WISH LIST FOR A BROADCAST CHANNEL OF RATE R BIT/S

1. When **one** node wants to transmit, it can send at rate R.
2. When M nodes want to transmit, each can send at **average** rate R/M.
3. **Fully decentralized**: no special node coordinates (no single point of failure).
4. **No synchronization** of clocks or slots.
5. **Simple**.

Keep this list in mind: every protocol below is scored against it in §14.1.

> 6.4 A taxonomy of MAC protocols



The lecture's two broad classes (slides 45, 89, 103).

- Channel partitioning: divide the channel into smaller "pieces" (time slots or frequencies) and give each node a piece for its exclusive use.
- Taking turns: nodes take turns, but nodes with more to send can take longer turns.
- Random access: the channel is not divided. Collisions are allowed, and the protocol *recovers* from them.

PITFALL · IS CHANNEL PARTITIONING "CENTRALIZED"?

The slides group channel partitioning under "Centralized Coordination". Once the slots or bands are assigned, TDMA and FDMA need *no* coordinator at run time: every node simply uses its own piece. What they need is an *agreement made in advance*. Only some taking-turns protocols (polling) have a real central master. That is why [KR 6.3] lists **three** classes side by side: channel partitioning, random access and taking turns. Also, slide 103 writes

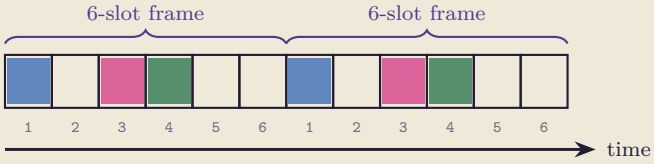
"time/frequency *modulation*". It should be time/frequency **division** (multiplexing). Modulation is how bits are put on a signal, a physical-layer matter.

SECTION 07

Channel partitioning: TDMA and FDMA

DEF · TDMA: TIME DIVISION MULTIPLE ACCESS

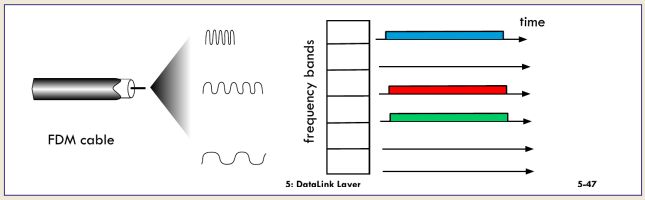
Access to the channel goes in **rounds** (frames). Each station gets a fixed-length slot in every round; the slot length is one packet transmission time. Unused slots go idle.



Slide 46's example: a 6-station LAN in which stations 1, 3 and 4 have packets. Slots 2, 5 and 6 stay idle in every frame, even though 1, 3 and 4 have more to send.

DEF · FDMA: FREQUENCY DIVISION MULTIPLE ACCESS

The channel's spectrum is divided into frequency bands, and each station is assigned a fixed band. Unused transmission time in a band goes idle.



Slide 47. One FDM cable carries six bands. Stations 1, 3, 4 send in their bands; bands 2, 5, 6 are idle.

EXAMPLE · THE COST OF PARTITIONING, WITH NUMBERS [KR 6.3.1]

A channel of $R = 60$ Mbps shared by $N = 6$ stations: each one gets exactly $R/N = 10$ Mbps.

- 6 stations busy: each gets 10 Mbps $= R/M$. Property 2 is met perfectly, and there are no collisions.
- Only station 1 busy: it still gets only 10 Mbps, and 50 Mbps are wasted. Property 1 fails.
- Even when alone, a TDMA station must wait for its own slot: on average half a frame of delay before each packet.

Like a pan of Kaiserschmarrn torn into six fixed portions before the guests arrive: if only one guest is hungry, they still get one sixth, and the rest goes cold.

SECTION 08

Random access protocols

DEF · RANDOM ACCESS

When a node has a packet to send, it transmits at the **full channel rate** R, with no prior coordination among the nodes. If two or more nodes transmit, there is a collision. A random access MAC protocol specifies:

- how to **detect** collisions;
- how to **recover** from them, e.g. by delayed retransmission.

Examples: slotted ALOHA, (pure) ALOHA, CSMA, CSMA/CD, CSMA/CA.

WHY THE DELAY MUST BE RANDOM

If two colliding nodes both waited the *same* fixed time and tried again, they would collide again, forever. Each node chooses its delay *independently at random*, so with luck one of them goes first and gets through [KR 6.3.2].

SECTION 09

Probability toolkit (recap of slides 49–67)

The lecture now pauses for "Probability 101": sample spaces, events, probability functions, conditional probability, Bayes, random variables, expectation, the four discrete distributions, and Examples I–III. **Week 01 Part B covers all of these in full**, including Example II (good and bad packets) and Example III (the chain of links). Here are only the facts that the MAC analysis below uses.

WHAT WE NEED

- **Independent events multiply**: $P(A \cap B) = P(A)P(B)$. So "$N-1$ independent nodes all stay silent, each with probability $1-p$" has probability $(1-p)^{N-1}$.
- **Disjoint events add**: $P(A \cup B) = P(A) + P(B)$.
- **Exactly one success among** N: $\binom{N}{1}p(1-p)^{N-1} = Np(1-p)^{N-1}$, the binomial probability $P(X=1)$ for $X \sim \mathrm{Bin}(N,p)$.
- **Geometric**: if each try succeeds with probability q, the number of tries is $\mathrm{Geo}(q)$, with mean $1/q$. The expected number of *failures* before the success is $1/q - 1 = (1-q)/q$.

EXAMPLE · EXAMPLE I FROM THE SLIDES: 50 COMPONENTS

Communication uses 50 components, each working 99% of the time (independently). Failure probability $= 1 - 0.99^{50} \approx 39.5\%$. With 99.9%: $1 - 0.999^{50} \approx 4.9\%$. The system must be

designed to expect failure.

PITFALL · SLIDE 67: "THE PROBABILITY TO SUCCESS DROPS TO $(1-p)$"

In Example III (Week 01, Question 3) the success probability goes from $(1-p)^n$ to $(1-p)$. Since $1-p > (1-p)^n$ for $n \geq 2$, it **rises**, not "drops". Also, the slide's $\mathbb{E}X = 1/(1-p)$ counts the sends needed to cross *one* hop. Crossing all n hops hop by hop takes $n/(1-p)$ link transmissions on average.

>9.1 The new tool: $(1-1/N)^N \to 1/e$

The efficiency of ALOHA ends with "take the limit as $N \to \infty$". This is the limit it needs.

A FAMOUS LIMIT

$$\lim_{N\to\infty}\left(1-\frac{1}{N}\right)^N = \frac{1}{e} \approx 0.368, \qquad \text{and therefore also} \qquad \lim_{N\to\infty}\left(1-\frac{1}{N}\right)^{N-1} = \frac{1}{e}.$$

(The second follows from the first: $(1-\frac{1}{N})^{N-1} = (1-\frac{1}{N})^N/(1-\frac{1}{N})$, and the denominator tends to 1.)

EXAMPLE · SEE IT CONVERGE

N	$(1-1/N)^N$	$(1-1/N)^{N-1}$
2	0.250	0.500
5	0.328	0.410
10	0.349	0.387
100	0.366	0.370
∞	0.368	0.368

Why it is true, in words: $\ln(1-x) \approx -x$ for small x, so $N\ln(1-\frac{1}{N}) \approx N\cdot(-\frac{1}{N}) = -1$, and $(1-\frac{1}{N})^N \approx e^{-1}$. In probability terms: if each of N nodes stays silent with probability $1-\frac{1}{N}$, then the chance that *all* stay silent tends to $1/e$. It is the Poisson probability $P(X=0)$ for mean 1.

SECTION 10

Slotted ALOHA

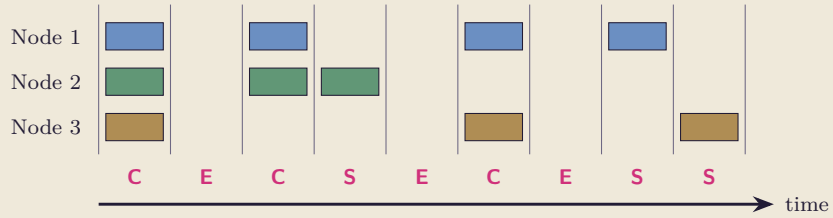
DEF · SLOTTED ALOHA

Assumptions: all frames have the same size; time is divided into equal slots, each the time to transmit one frame; nodes start transmitting only at the beginning of a slot; nodes are synchronized; if two or more nodes transmit in a slot, all nodes detect the collision.

Operation:

1. When a node gets a fresh frame, it transmits in the next slot.
2. No collision: success; the node may send a new frame in the next slot.
3. Collision: the node retransmits the frame in *each subsequent slot with probability* p, until it succeeds.

"With probability p": before each slot the node tosses a biased coin, and transmits on heads (probability p).



Slotted ALOHA in action [KR Fig. 6.10]. C = collision, E = empty, S = success. All three nodes collide in slot 1. Node 2 succeeds in slot 4, node 1 in slot 8 and node 3 in slot 9.

Pros: a single active node can transmit continuously at the full rate R; highly decentralized (only the slot times must be in sync); simple.
Cons: collisions waste slots; idle slots are wasted too; a node may detect a collision long before the slot ends but still wastes the whole slot; it needs clock synchronization.

> 10.1 Goodput vs. throughput

DEF · GOODPUT AND THROUGHPUT

Consider many nodes, all with many frames to send.

- Goodput (also efficiency): the long-run fraction of *successful* slots.
- Throughput: the long-run fraction of slots in which there was a transmission, successful or not.

Throughput counts collisions as "busy"; goodput counts only useful work. A channel full of collisions has high throughput and zero goodput.

> 10.2 The goodput of slotted ALOHA, step by step

Simplified model. N nodes, each always has a frame, and in each slot *each* node transmits with probability p, independently (this treats fresh frames and retransmissions alike).

1. **A given node succeeds** in a slot iff it transmits *and* the other $N-1$ do not. By independence:

$$P(\text{node } i \text{ succeeds}) = p\,(1-p)^{N-1}.$$

2. **Some node succeeds**: the events "node i succeeds" for $i = 1, \ldots, N$ are *disjoint* (two nodes cannot both succeed in one slot), so we add:

$$\alpha(p) = N\,p\,(1-p)^{N-1}.$$

3. **Best** p: differentiate and set to zero (slide 84):

$$\frac{d\alpha}{dp} = N(1-p)^{N-1} - pN(N-1)(1-p)^{N-2} = N(1-p)^{N-2}\left[(1-p)-(N-1)p\right] = 0 \;\Rightarrow\; p^* = \frac{1}{N}.$$

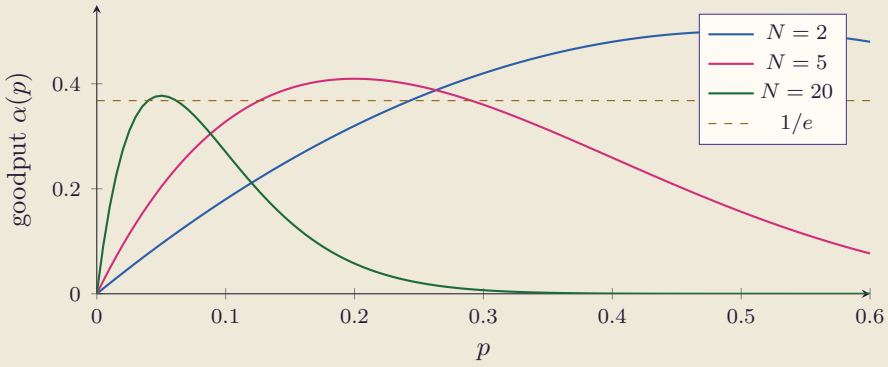
   Makes sense: on average exactly one node transmits per slot.

4. **Best goodput**: $\alpha(p^*) = N \cdot \frac{1}{N}\left(1-\frac{1}{N}\right)^{N-1} = \left(1-\frac{1}{N}\right)^{N-1}$.

5. **Many nodes**: by the limit above, $\alpha_{\max} \to \dfrac{1}{e} \approx 0.37$.

PITFALL · SLIDE 73: "APPROX." $Np(1-p)^{N-1}$

The slide calls $Np(1-p)^{N-1}$ "approximately" the probability that some node succeeds. Within the model it is **exact**: the N success events are disjoint, so their probabilities add with no approximation. The approximation comes only in the last step, when $(1-\frac{1}{N})^{N-1}$ is replaced by its limit $1/e$.



Slotted ALOHA goodput as a function of p. Each curve peaks at $p = 1/N$; the peak heights 0.50, 0.41, 0.38 approach $1/e$.

EXAMPLE · WHERE THE SLOTS GO [KR 6.3.2]

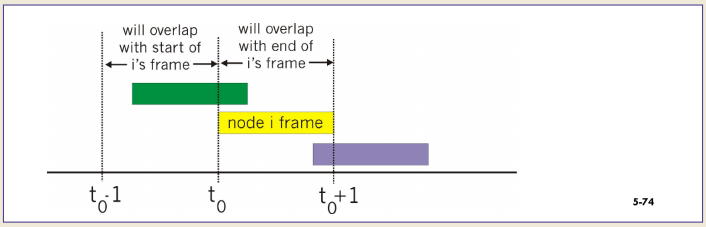
With many nodes at $p = 1/N$: 37% of slots succeed, 37% are empty ($(1-\frac{1}{N})^N \to 1/e$), and the remaining 26% are collisions. A 100 Mbps slotted ALOHA LAN with many busy users delivers less than 37 Mbps of useful data.

SECTION 11

Pure (unslotted) ALOHA

DEF · PURE ALOHA

No slots and no synchronization: simpler. Each host behaves *as if* it ran slotted ALOHA on its *own* slots, sending a frame with probability p in each of them. But the other hosts' "slots" are not aligned with its own.



Slide 74. Node i's frame (yellow) occupies $[t_0, t_0 + 1]$ (the frame time is the unit of time). A frame started in $[t_0 - 1, t_0]$ (green) overlaps its start; a frame started in $[t_0, t_0 + 1]$ (purple) overlaps its end.

Step by step.

1. The vulnerable period is **two** frame times, $[t_0 - 1, t_0 + 1]$, instead of one.
2. $P(\text{node } i \text{ succeeds}) = P(\text{it transmits}) \cdot P(\text{no other in } [t_0 - 1, t_0]) \cdot P(\text{no other in } [t_0, t_0 + 1]) = p\,(1-p)^{N-1}(1-p)^{N-1} = p\,(1-p)^{2(N-1)}$.
3. Goodput $Np(1-p)^{2(N-1)}$. The best p is $p^* = 1/(2N-1)$, and as $N \to \infty$ the maximum tends to $\dfrac{1}{2e} \approx 0.18$.

HALF OF SLOTTED ALOHA

Pure ALOHA's maximum is exactly half of slotted ALOHA's: 0.18 vs. 0.37. The vulnerable window is twice as long. That is the price of not synchronizing clocks.

N	SLOTTED, $(1-\frac{1}{N})^{N-1}$	PURE, at $p^* = \frac{1}{2N-1}$
2	0.500	0.296
5	0.410	0.216
20	0.377	0.191
∞	$1/e = 0.368$	$1/(2e) = 0.184$

SECTION 12

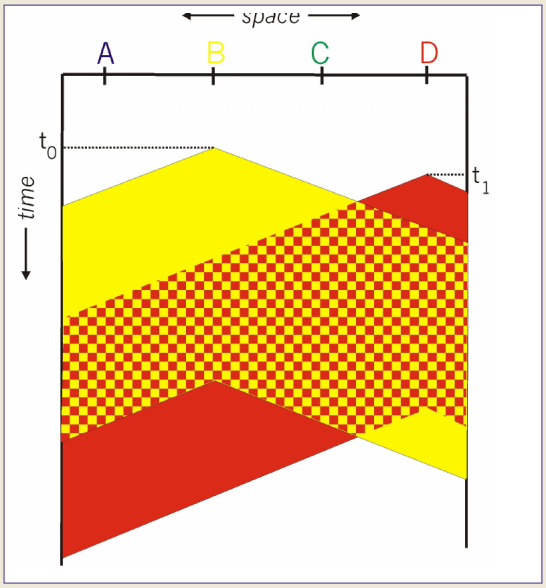
CSMA: listen before you talk

DEF · CSMA (CARRIER SENSE MULTIPLE ACCESS)

- If the channel is sensed **idle**: transmit the entire frame.
- If the channel is sensed **busy**: defer the transmission.

Human analogy: *don't interrupt others!* ALOHA, in contrast, is the guest who keeps talking no matter who else is speaking.

> 12.1 Why CSMA still has collisions



Slide 77, a space-time diagram: space horizontally (four nodes on a bus), time downwards. B starts at t_0; its signal (yellow) spreads in both directions. At t_1 B's signal has not yet reached D, so D senses an idle channel and starts too (red). The chequered area is where they collide.

THE ROLE OF PROPAGATION DELAY

Carrier sensing only tells a node what was on the channel *one propagation delay ago*. Two nodes may each sense "idle" while the other's signal is still on its way. The longer the propagation delay between nodes, the larger the collision probability. In plain CSMA, when a collision happens, **the whole frame transmission time is wasted**: B and D both keep transmitting to the end.

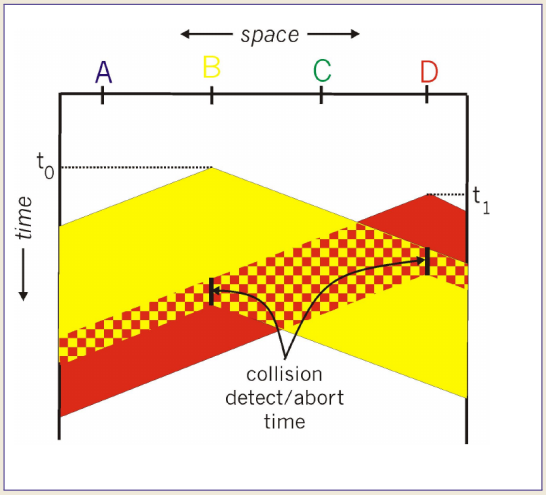
SECTION 13

CSMA/CD: collision detection

DEF · CSMA/CD

Carrier sensing as in CSMA, plus collision detection: a node keeps listening while it transmits. Collisions are detected within a short time, and the colliding transmissions are **aborted**, which reduces the waste. Human analogy: the polite conversationalist who stops when someone else starts talking.

- **Easy in wired LANs**: compare the signal transmitted with the signal received.
- **Hard in wireless LANs**: the node's own transmission overwhelms the weak signal it receives from others.



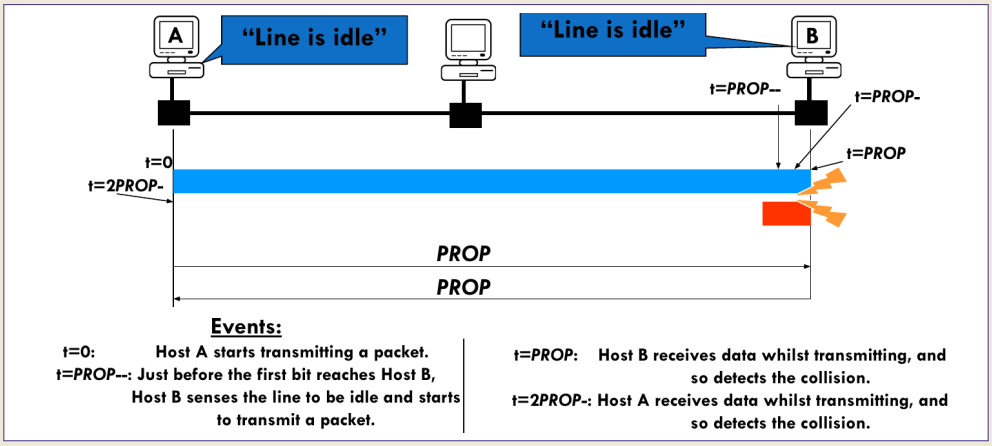
Slide 79: the same scenario with collision detection. B and D each stop shortly after the other's signal arrives. The wasted area is much smaller than in plain CSMA.

THE CSMA/CD PROTOCOL (SLIDE 80)

All hosts transmit and receive on one channel. When a host has a packet to send:

1. **Carrier sense**: check that the line is quiet before transmitting.
2. **Collision detection**: detect a collision as soon as possible. If one is detected, stop transmitting, wait a random time, and return to step 1.

> 13.1 The worst case: how long until a collision is detected?



Slide 81. A and B are at the two ends of the cable, one propagation delay PROP apart.

The events, step by step (ε is a tiny time):

1. $t = 0$: A senses idle and starts transmitting.
2. $t = \text{PROP} - \varepsilon$: A's first bit has *almost* reached B. B senses the line idle and starts its own packet.
3. $t = \text{PROP}$: A's bits reach B. B is receiving while transmitting, so **B detects the collision** (after only ε).
4. B's first bits travel back to A, arriving at $t = 2\,\text{PROP} - \varepsilon$. Only now does **A detect the collision**.

MINIMUM FRAME DURATION (SLIDE 82)

To be sure that a packet was sent without collision, a host must be able to detect a collision **before it finishes transmitting** the packet. In the worst case detection takes almost 2 PROP, so

$$\text{TRANSP} \;\geq\; 2 \cdot \text{PROP}.$$

If A had already finished at $t < 2\,\text{PROP}$, it would wrongly believe the packet had got through.

EXAMPLE · WHY ETHERNET WAITS IN UNITS OF 512 BITS

Classic 10 Mbps Ethernet uses 512 bit times as its basic unit (slide 91). At 10 Mbps one bit takes 0.1 μs, so 512 bits take 51.2 μs. A frame of at least 512 bits (64 bytes) therefore lasts at least 51.2 μs, which covers a round trip 2 PROP of up to 51.2 μs. At 2×10^8 m/s that is about 5 km of round trip, so about 2.5 km of cable end to end (in practice less, because repeaters add delay).

> 13.2 The goodput of CSMA/CD (slides 83-88)

DEF · EFFICIENCY η

$$\eta = \frac{\text{time taken to send data}}{\text{time taken to send data} + \text{overhead}}.$$

Model. Time is slotted, and all packets have the same length. A **slot is 2 PROP**, the longest it can take to detect a collision. In each slot, each of N hosts decides to transmit with probability p (first transmissions and retransmissions alike).

1. **A slot is "won"** when exactly one host transmits. As in slotted ALOHA,

$$\alpha(p) = \binom{N}{1} p(1-p)^{N-1}, \qquad \alpha_{\max} = \left(1 - \tfrac{1}{N}\right)^{N-1} \text{ at } p = \tfrac{1}{N}.$$

2. **Wasted slots.** Let A be the expected number of slots wasted before a packet gets through. In the first slot: with probability α we win and waste 0; with probability $1-\alpha$ we waste this slot and are back where we started, still expecting A more:

$$A = \alpha \cdot 0 + (1-\alpha)(1+A) \quad \Rightarrow \quad A = \frac{1-\alpha}{\alpha}.$$

   This is exactly "failures before the first success" of a geometric variable.

3. **With** $\alpha = 0.4$: $A = 0.6/0.4 = 1.5$. Coin check (slide 85): a coin with $P(\text{heads}) = 0.4$ needs on average $1/0.4 = 2.5$ tosses, i.e. 1.5 failures and then 1 success.

4. **Efficiency.** Each packet costs its transmission time plus A wasted slots of 2 PROP:

$$\eta_{\text{CSMA/CD}} = \frac{\text{TRANSP}}{\text{TRANSP} + A \cdot 2\,\text{PROP}} = \frac{\text{TRANSP}}{\text{TRANSP} + 3\,\text{PROP}} = \frac{1}{1+3a}, \qquad a \equiv \frac{\text{PROP}}{\text{TRANSP}}.$$

5. **More precise models and simulation** give $\eta_{\text{CSMA/CD}} \approx \dfrac{1}{1+5a}$.

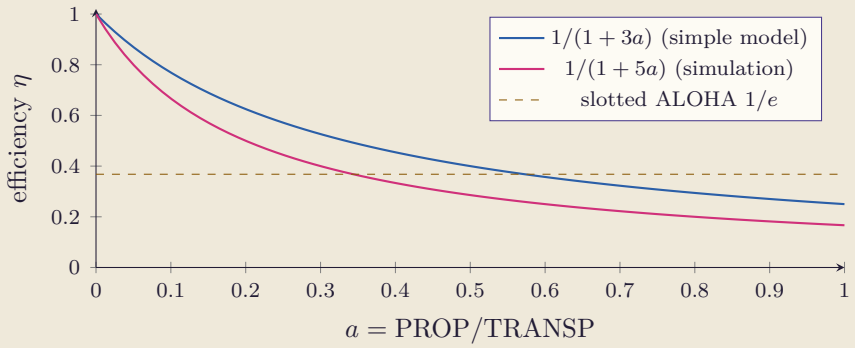
PITFALL · SLIDE 84: "$\alpha_{\max} \approx 37\% \approx 40\%$"

37% and 40% are not the same number. $\alpha_{\max} = (1 - \frac{1}{N})^{N-1}$ is 41% for $N = 5$ and falls to $1/e = 37\%$ only as $N \to \infty$. The slides then round to $\alpha = 0.4$ to get the neat $A = 1.5$. With the large-N value $\alpha = 1/e$ we get $A = e - 1 \approx 1.72$ and $\eta = 1/(1 + 3.44a)$. The message is the same: the overhead is a small multiple of PROP per packet.

EVERYTHING DEPENDS ON a = PROP/TRANSP

a compares "how long it takes to find out about a collision" with "how long the useful work lasts".

- $a \to 0$ (short cable or long frames): $\eta \to 1$. A collision is noticed and aborted almost instantly.
- a large (long distances, very fast links, short frames): η collapses. This is why CSMA/CD only works in a *local* area.



CSMA/CD efficiency vs. a. For a LAN, a is around 0.01, so η is above 90%, far better than ALOHA's 37%.

SECTION 14

Summary of MAC protocols

- **Channel partitioning** by time, frequency or code: TDMA, FDMA (and CDMA).
- **Taking turns**:
  - polling from a central site: a master node invites each node in turn to send up to some maximum number of frames;
  - token passing: a special "token" frame is passed from node to node, and only the holder may send.

  Examples: Bluetooth, FDDI, IBM Token Ring [KR 6.3.3].
- **Random access**: ALOHA, slotted ALOHA, CSMA, CSMA/CD. Carrier sensing is easy in some technologies (wire) and hard in others (wireless). CSMA/CD is used in Ethernet; CSMA/CA in 802.11 (WiFi).

> 14.1 Scorecard against the wish list

PROTOCOL	1: ALONE $\to R$	2: M BUSY $\to R/M$	3: DECENTR.	4: NO SYNC	5: SIMPLE
TDMA	no (R/N)	yes	yes*	no	yes
FDMA	no (R/N)	yes	yes*	yes	yes
Polling	almost	almost	no (master)	yes	fairly
Slotted ALOHA	yes	no ($\leq 0.37R$)	yes	no	yes
Pure ALOHA	yes	no ($\leq 0.18R$)	yes	yes	yes
CSMA/CD	yes	nearly ($\eta \approx \frac{1}{1+5a}$)	yes	yes	yes

* once the slots or bands have been assigned. No protocol scores "yes" everywhere; random access gives up property 2 under heavy load in exchange for everything else.

SECTION 15

Ethernet's CSMA/CD algorithm

ETHERNET CSMA/CD, AS RUN BY THE NIC (SLIDE 90)

1. The NIC receives a datagram from the network layer and creates a frame.
2. If the NIC senses the channel **idle**, it starts transmitting the frame. If it senses the channel **busy**, it waits until the channel is idle, then transmits.
3. If the NIC transmits the entire frame without detecting another transmission, it is done with the frame.
4. If the NIC detects another transmission while transmitting, it **aborts** and sends a jam signal.
5. After aborting, the NIC enters binary exponential backoff: after the m-th collision it chooses K at random from $\{0, 1, 2, \ldots, 2^m - 1\}$, waits $K \cdot 512$ bit times, and returns to step 2.

DEF · JAM SIGNAL AND BIT TIME

- Jam signal: 48 bits sent after aborting, to make sure that *all* other transmitters notice the collision.
- Bit time: the time to send one bit: $0.1\,\mu$s at 10 Mbps (and $0.01\,\mu$s at 100 Mbps).

> 15.1 Exponential backoff, step by step

Goal: adapt the retransmission attempts to the current load. Under heavy load, the random wait should be longer.

COLLISION m	K CHOSEN FROM	MAX WAIT AT 10 MBPS	P(2 NODES PICK SAME K)
1	$\{0, 1\}$	$51.2\,\mu$s	$1/2$
2	$\{0, \ldots, 3\}$	$153.6\,\mu$s	$1/4$
3	$\{0, \ldots, 7\}$	$358.4\,\mu$s	$1/8$
10 and up	$\{0, \ldots, 1023\}$	≈ 52.4 ms	$1/1024$

- The wait for K is $K \cdot 512 \cdot 0.1\,\mu\text{s} = K \cdot 51.2\,\mu$s. For $K = 1023$: $1023 \times 51.2\,\mu\text{s} \approx 52.4$ ms, the slide's "about 50 msec".
- The last column is for *two* colliding nodes: they collide again only if they pick the same K, which happens with probability $1/2^m$.

WHY "EXPONENTIAL" IS THE RIGHT SHAPE [KR 6.3.2]

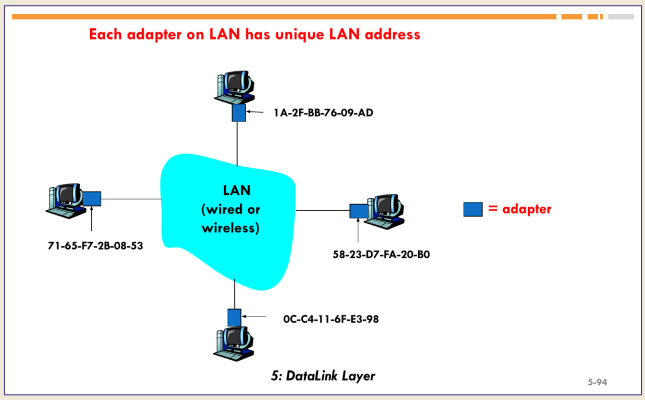
We want the interval to be **short when few nodes collide** (so the channel is not left idle) and **long when many collide** (so their random choices spread out). Nobody knows how many nodes are colliding, but each new collision is evidence that there are many, so the interval doubles. The doubling stops at $m = 10$.

SECTION 16

MAC addresses

DEF · MAC ADDRESS

A MAC address (also LAN, physical or Ethernet address) is used to get a frame from one interface to another *physically connected* interface on the same network. For most LANs it is **48 bits** long, written as 6 bytes in hexadecimal, e.g. `1A-2F-BB-76-09-AD`. It is burned into the NIC's ROM, and is sometimes also settable in software.



Slide 94. Each adapter (blue) on the LAN has its own unique MAC address. Addresses belong to adapters, not to hosts: a host with two NICs has two MAC addresses [KR 6.4.1].

WHO HANDS OUT MAC ADDRESSES

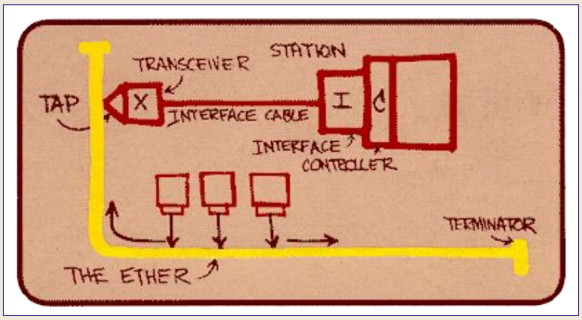
The IEEE administers the address space. A manufacturer buys a block of addresses: the IEEE fixes the first 24 bits, and the manufacturer assigns the last 24 bits, which is $2^{24} \approx 16.8$ million unique addresses per block [KR 6.4.1]. In total there are $2^{48} \approx 2.8 \times 10^{14}$ addresses.

FLAT VS. HIERARCHICAL: ID NUMBER VS. POSTAL ADDRESS

A MAC address is like your **ID number** (*teudat zehut*): a flat identifier that says nothing about where you are, and stays the same wherever you go. So a network card can be moved from one LAN to another and keep its address: flat ⇒ portable. An IP address (later in the course) is hierarchical, like a postal address: it encodes *where* you are, so it must change when you move, and it is not portable.

SECTION 17

Ethernet



Bob Metcalfe's original Ethernet sketch (slide 97): stations attached through transceivers to one shared cable, "the ether".

DEF · ETHERNET

The "dominant" wired LAN technology: cheap (about $20 for a NIC); the first widely used LAN technology; simpler and cheaper than token LANs and ATM; it kept up with the speed race, from 10 Mbps to 100 Gbps. Metcalfe created it by modifying the ALOHA protocol into CSMA/CD [KR 6.3.2].

ETHERNET GOODPUT (SLIDE 98)

Let t_{prop} = the maximum propagation delay between two nodes in the LAN, and t_{trans} = the time to transmit a maximum-size frame. Then

$$\text{goodput} \approx \frac{1}{1 + 5\, t_{\text{prop}}/t_{\text{trans}}}.$$

Goodput tends to 1 as $t_{\text{prop}} \to 0$ and as $t_{\text{trans}} \to \infty$. This is the $1/(1+5a)$ of CSMA/CD with $a = t_{\text{prop}}/t_{\text{trans}}$.

EXAMPLE · A 10 MBPS ETHERNET WITH NUMBERS

Maximum frame $\approx$ 1518 bytes = 12,144 bits, so $t_{\text{trans}} = 12{,}144/10^7 \approx 1.21$ ms. A 500 m cable at 2×10^8 m/s: $t_{\text{prop}} = 2.5\,\mu$s. Then $a = 2.5/1214 \approx 0.0021$ and goodput $\approx 1/(1 + 0.0103) \approx 99\%$.

Now speed the same LAN up to 1 Gbps: $t_{\text{trans}} \approx 12.1\,\mu$s, $a \approx 0.21$, goodput $\approx 1/(1{+}1.03) \approx 49\%$. Faster links shrink t_{trans} and make the shared-bus design worse, one reason today's Ethernet uses switched stars.

SECTION 18

WiFi (IEEE 802.11): CSMA/CA

WHY WIFI CANNOT USE COLLISION DETECTION (SLIDE 99)

802.11 does carrier sensing (it does not start while another node is transmitting), but it has **no collision detection**:

- it is hard to receive (sense a collision) while transmitting, because the received signals are weak compared to the node's own (fading);
- even then, it could not sense all collisions: the **hidden terminal** problem and fading mean the collision happens at the receiver, out of the sender's hearing.

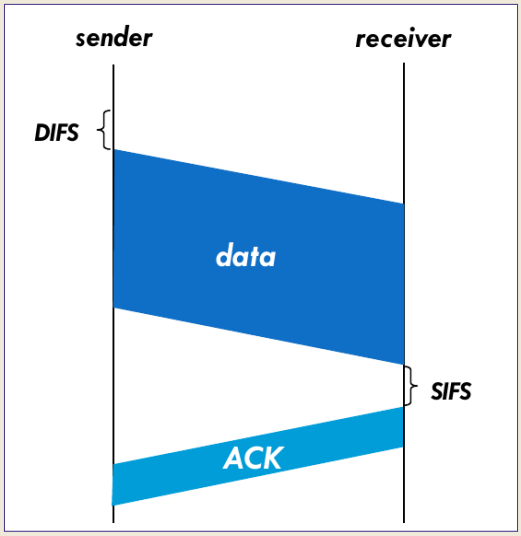
So the goal changes: **avoid** collisions. Hence CSMA/C(ollision)A(voidance).

DEF · 802.11 CSMA/CA (SLIDE 100)

Sender:

1. If it senses the channel idle for a period called DIFS (distributed inter-frame space), it transmits the entire frame (there is no CD, so there is no aborting).
2. If it senses the channel busy, it starts a **random backoff** timer. The timer counts down only while the channel is idle (it is frozen while busy). The node transmits when the timer expires. If no ACK comes back, it increases the random backoff interval and repeats step 2.

Receiver: if the frame is received OK, it returns an ACK after a short period SIFS (short inter-frame space).



Slide 100. The sender waits DIFS, sends the data; the receiver waits SIFS and sends the ACK.

WHY ACKS, AND WHY BACK OFF EVEN AFTER AN IDLE CHANNEL [KR 7.3.2]

- **ACKs**: without collision detection, the sender cannot know whether its frame arrived. Wireless links also have high bit-error rates. The link-layer ACK is the only evidence.
- **Backoff**: suppose two stations wait for a third to finish. With Ethernet's rule they would both start the moment it ends, and collide, which is cheap in Ethernet because they abort. In WiFi a collision costs the *whole* frame. So both draw random backoffs instead; the one with the smaller value goes first, and the other hears it and freezes its counter.

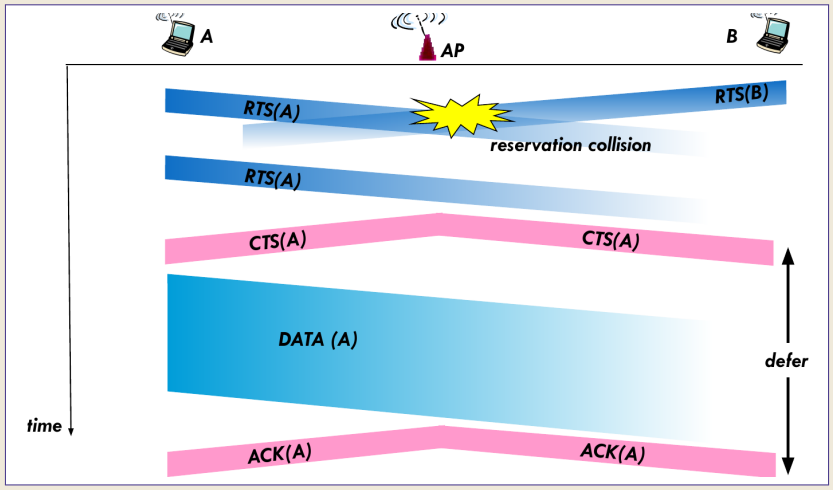
> 18.1 Reserving the channel: RTS and CTS

DEF · RTS/CTS (SLIDE 101)

Idea: let the sender **reserve** the channel instead of sending long data frames at random.

1. The sender first transmits a small request-to-send (RTS) packet to the access point (AP), using CSMA. RTSs may still collide with each other, but they are short, so little time is lost.
2. The AP broadcasts a clear-to-send (CTS) in response to the RTS.
3. **Every node hears the CTS**, including nodes hidden from the sender. The sender transmits its data frame, and the other stations defer.

Collisions of data frames are avoided completely, using small reservation packets.



Slide 102. A and B send RTS at the same time: a *reservation collision* (cheap, the RTS is short). A retries; the AP answers CTS(A), heard by both A and B. A sends DATA(A); B defers until the ACK(A).

EXAMPLE · WHY RTS/CTS SOLVES THE HIDDEN TERMINAL

Take slide 43's picture: A and C are hidden from each other behind the mountain, and both reach B, the AP.

1. A sends an RTS to B. C does not hear it.
2. B broadcasts CTS(A). **C hears the CTS**, because C does reach B.

3. C now knows the channel is reserved, and stays quiet until it hears B's ACK(A).

Carrier sensing failed because C could not hear A, the *sender*. RTS/CTS works because C can hear B, the *receiver*, which is exactly where collisions happen.

EXAMPLE · IS THE RESERVATION WORTH IT?

Say RTS and CTS take 20 μs each and a data frame takes 1000 μs. A data-frame collision wastes 1000 μs; an RTS collision wastes only 20 μs. The price is about 40 μs of extra control per frame, 4%. It pays off when collisions are likely. It does not pay off for very short frames, where the RTS/CTS would be as long as the data.

SECTION 19

Summary: MAC protocols in practice

WEEK 02 IN ONE PARAGRAPH

Networks are built as a stack of five layers: physical, link, network, transport, application. Each layer talks only to its neighbours and its peers, and adds its own header (encapsulation). IP is the single narrow waist. Building bottom up, the first challenge is a LAN: a shared broadcast link needs a MAC protocol. **Channel partitioning** (TDMA, FDMA) is fair but wastes the channel when few nodes are busy. **Random access** gives a lone node the full rate but loses to collisions: slotted ALOHA at most $1/e \approx 37\%$, pure ALOHA $1/(2e) \approx 18\%$. Listening first (CSMA) and aborting early (CSMA/CD, *Ethernet*) push the efficiency to $1/(1+5a)$, close to 100% in a LAN. Wireless cannot detect collisions, so *802.11* avoids them (CSMA/CA, ACKs, RTS/CTS).

WEEK 02

Practice Problems and Summary

SECTION 20

Practice problems with full solutions

EXAMPLE · P1: LAYERS

A router receives an Ethernet frame and forwards the packet on a SONET link. Which headers does it read, and which does it replace?

Solution. It removes the Ethernet (link) header, reads the IP (network) header to choose the outgoing link, decrements the TTL, and wraps the datagram in a new SONET link header. It never reads the TCP or HTTP headers: transport and application exist only at the hosts.

EXAMPLE · P2: ENCAPSULATION OVERHEAD

An application sends 100 bytes over TCP/IP/Ethernet. What fraction of the frame (headers only) is data? And for 1000 bytes?

Solution. $100/154 = 64.9\%$. $1000/1054 = 94.9\%$.

EXAMPLE · P3: SLOTTED ALOHA

$N = 4$ nodes, each transmits in a slot with probability $p = 0.2$. (a) Probability that node 1 succeeds in a slot? (b) Goodput? (c) The best p and its goodput?

Solution. (a) $0.2 \cdot 0.8^3 = 0.1024$. (b) $4 \times 0.1024 = 0.4096$. (c) $p^* = 1/4$: $(3/4)^3 = 0.4219$.

EXAMPLE · P4: EMPTY SLOTS

In P3 with $p = 0.2$, what fraction of slots is empty, and what fraction has collisions?

Solution. Empty: $0.8^4 = 0.4096$. Collisions: $1 - 0.4096 - 0.4096 = 0.1808$.

EXAMPLE · P5: PURE ALOHA

$N = 3$ nodes, $p = 0.2$. Goodput of pure ALOHA? Compare with slotted ALOHA.

Solution. Pure: $3 \cdot 0.2 \cdot 0.8^4 = 0.2458$. Slotted: $3 \cdot 0.2 \cdot 0.8^2 = 0.384$. Pure is lower because the vulnerable window is twice as long.

EXAMPLE · P6: CSMA/CD MINIMUM FRAME

A CSMA/CD LAN runs at 100 Mbps over at most 200 m of cable ($s = 2 \times 10^8$ m/s). What is the minimum frame size?

Solution. PROP $= 200/(2 \times 10^8) = 1\,\mu$s. Need TRANSP $\geq 2\,$PROP $= 2\,\mu$s, so $L \geq 2 \times 10^{-6} \times 10^8 = 200$ bits $= 25$ bytes.

EXAMPLE · P7: CSMA/CD EFFICIENCY

PROP $= 5\,\mu$s and TRANSP $= 100\,\mu$s. Find a and the efficiency in both models.

Solution. $a = 0.05$. Simple model: $1/(1+0.15) = 87\%$. Simulation model: $1/(1+0.25) = 80\%$.

EXAMPLE · P8: WASTED SLOTS

In the CSMA/CD model, a slot is won with probability $\alpha = 0.25$. Expected wasted slots per packet?

Solution. $A = (1-\alpha)/\alpha = 0.75/0.25 = 3$ slots, i.e. $3 \times 2\,$PROP $= 6\,$PROP of overhead per packet.

EXAMPLE · P9: EXPONENTIAL BACKOFF

Two Ethernet NICs collide for the 3rd time on a 10 Mbps LAN. (a) From which set is K chosen? (b) Probability that they collide again? (c) Longest possible wait?

Solution. (a) $\{0, \ldots, 7\}$. (b) They pick the same K with probability $8 \cdot (1/8)^2 = 1/8$. (c) $7 \times 512 \times 0.1\,\mu$s $= 358.4\,\mu$s.

EXAMPLE · P10: HIDDEN TERMINAL

A, B and C are in a line; A and C are each 80 m from B, and the radio range is 100 m. Can A and C hear each other? Which mechanism stops their data frames from colliding at B?

Solution. A and C are 160 m apart, beyond the range: they are hidden from each other, so carrier sensing fails. RTS/CTS: B's CTS reaches both, so the station that did not win the CTS defers.

SECTION 21

Cheat sheet

WEEK 02 ON ONE PAGE

- Modularity: interfaces hide implementation; extensible; can cost performance. Network design choices: layering, where to implement, where to keep state.
- Layers: Physical (bits) / Link (frames, local addresses, MAC) / Network (packets, global addresses, routing, IP only) / Transport (processes; TCP, UDP) / Application. Routers: bottom 3. Hosts: all 5.
- Protocol = syntax (format, order) + semantics (meaning, actions). Standards: IETF RFCs. Layer = service + interface + protocol.
- Encapsulation: each layer adds a header. TCP 20 + IP 20 + Ethernet 14 = 54 bytes.

Hourglass: IP is the narrow *waist*.

- MAC protocol: distributed; coordinates over the channel itself. Wish list: alone $\to R$; M busy $\to R/M$; decentralized; no sync; simple.
- TDMA/FDMA: no collisions, fair, but R/N even when alone; unused slots/bands idle.
- Slotted ALOHA: $\alpha = Np(1-p)^{N-1}$, $p^* = 1/N$, max $(1-\frac{1}{N})^{N-1} \to 1/e \approx 0.37$. Pure ALOHA: $Np(1-p)^{2(N-1)}$, max $\to 1/(2e) \approx 0.18$.
- CSMA: listen first; collisions still happen because of propagation delay. CSMA/CD: abort on collision; need TRANSP $\geq 2\,$PROP.
- CSMA/CD goodput: $A = (1-\alpha)/\alpha$; $\eta = 1/(1+3a)$ (simple), $\approx 1/(1+5a)$ (precise), $a =$ PROP/TRANSP.
- Ethernet: after the m-th collision $K \in \{0, \ldots, 2^m - 1\}$ (capped at $m = 10$), wait $K \cdot 512$ bit times; jam 48 bits; bit time $0.1\,\mu$s at 10 Mbps.
- MAC address: 48 bits, flat, portable, IEEE gives 24-bit prefixes. IP address: hierarchical, not portable.
- 802.11: no CD (weak received signal, hidden terminal, fading) $\Rightarrow$ CSMA/CA: DIFS, random backoff frozen while busy, ACK after SIFS; RTS/CTS reserves the channel.